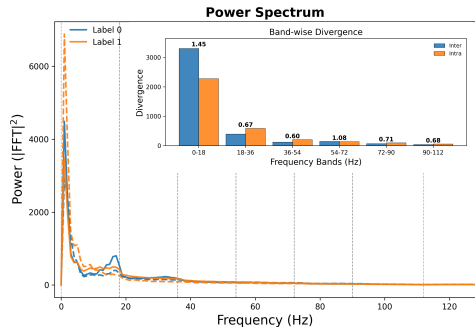


$$\mathcal{P} = \{(i, j) \mid y_i = y_j, s_i \neq s_j\}$$

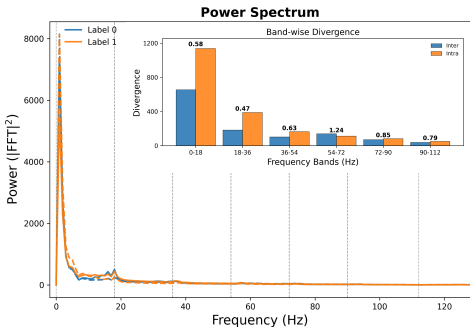
$$\mathcal{L}_{\text{band-align}} = \sum_{k=1}^K \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \left\| z_i^{(k)} - z_j^{(k)} \right\|_2^2$$

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{band}} \mathcal{L}_{\text{band-align}}$$

$z_i^{(k)}$ denotes the representation of sample i in the k -th band, and the loss is computed over all label-consistent but subject-discrepant pairs within each mini-batch, involving all eligible samples.



(a) w/o $\mathcal{L}_{\text{band-align}}$



(b) w/ $\mathcal{L}_{\text{band-align}}$